

Ömer Çayır

DATA ENGINEER · REAL-TIME SYSTEMS & BACKEND SPECIALIST

ISTANBUL

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Education

Karadeniz Technical University

Computer Engineering

Trabzon

2017 – 2023

- KTU Artificial Intelligence Community Founding Member - Former Board Member

Experience

IGA Istanbul Airport

Data Engineer

Istanbul

Mar 2023 – Present

- Managing enterprise-level data architectures, ensuring fault tolerance and low latency for mission-critical operational data flows.
- Utilized Apache Spark to transform a 15-minute data latency into real-time processing for critical flight operations.
- Orchestrating Airflow ETL/ELT processes to ensure scalability and efficiency for corporate BI units.
- Optimizing system performance by provisioning containerized environments and implementing advanced Redis caching.
- Developing full-stack operational tools to bridge the gap between complex data infrastructures and real-time user insights.

TÜBİTAK

STAR Scholar – TÜBİTAK 1001

Trabzon

May 2022 – Jan. 2023

- Developed dynamic G-Code generation algorithms and data visualization tools for industrial processes using Python.
- Conducted data analysis and visualization on large datasets using Pandas, NumPy, and Matplotlib.

Bordo Bilişim Ticaret Ltd. Şti.

Web Development Intern

Samsun

Sep 2021 – May 2022

- Actively participated in the development of reactive web applications using the MeteorJS and JavaScript ecosystem.
- Worked on NoSQL (MongoDB) database structures and frontend modernization using Bootstrap 5.

Projects

FIRST – Real-Time Operational Decision Support Platform

IGA Istanbul Airport

Istanbul

May 2024 – Present

- Developed a mission-critical platform using MERN Stack, creating custom UI components to visualize real-time aircraft status and planning.
- Architected a high-throughput backend using Apache Kafka and Spark to process live AMS streams, achieving sub-second latency.
- Streamlined data availability for operational teams, replacing legacy delayed reporting with instantaneous data visualization.

Unified Real-Time & Batch Integration Framework

IGA Istanbul Airport

Istanbul

Jun 2023 – Present

- Designed a robust framework to synchronize disparate data sources, including ASMGCS, AMS, SAP, and MSSQL, into a centralized environment.
- Developed ingestion layers for real-time and batch workloads, ensuring a "single source of truth" for enterprise reporting.
- Optimized data movement via Redis and Spark, delivering robust and high-performance analytical solutions.
- Modernized legacy data pipelines to handle real-time AMS data, reducing data availability latency from 15 minutes to sub-second intervals, significantly enhancing the responsiveness of decision support systems.

Enterprise Data Warehouse & ETL Architecture

IGA Istanbul Airport

Istanbul

Jun 2023 – Present

- Engineered a multi-layered data architecture consisting of Staging (STG), Operational Data Store (ODS), and Data Warehouse (DWH) layers.
- Implemented complex transformation logic to create a "single source of truth," enabling high-level executive analytics and operational reporting.
- Established data quality protocols at each layer to ensure 100% data integrity across PostgreSQL and MongoDB repositories.

Skills

Data Engineering

Apache Kafka, Apache Spark, Apache Airflow, Redis (Caching), ETL/ELT Pipelines

Databases & Arch.

PostgreSQL (Advanced), MongoDB, MSSQL, Multi-layered DWH (STG, ODS, DWH)

Software Development

JavaScript, Node.js, ReactJS, MERN Stack, Python, C/C++, Shell Scripting (Bash/Zsh)

Infrastructure & Tools

Docker, Kubernetes, Linux, Git, LaTeX, Data Visualization (Heatmaps)

Certificates

Docker	Gantek Academy	Jun 2024
Kubernetes	Gantek Academy	Feb 2025
Apache Kafka	Gantek Academy	Feb 2025

Languages

Native Language: Turkish

English	B1 (Intermediate)
Arabic	A1 (Elementary)